INC-2026-03-14: Payment authorisation failures following connection pool change

Northwind Payments · SEV1 · service payment-gateway · author s.petrova

Started 2026-03-14T14:32:00Z · detected 2026-03-14T14:38:00Z · resolved 2026-03-14T15:10:00Z

Customer impact

Approximately 40% of checkout authorisation requests returned HTTP 503 for 38 minutes. An estimated 11,400 payment attempts failed. Merchants saw elevated decline rates and the merchant portal transaction view returned partial data.

Root cause

Release pay-gw-4.2.1 reduced the per-pod database connection pool from 50 to 10. Under normal afternoon load the gateway requires roughly 28 concurrent connections per pod, so the pool was exhausted within twelve minutes of the rollout completing. Exhausted checkouts waited on pool acquisition until the 2s upstream timeout fired and returned 503.

Causing change

This incident was caused by release pay-gw-4.2.1, deployed to payment-gateway. The release is recorded in the change log.

Affected services

payment-gateway, checkout-api, merchant-portal. Downstream services were affected through their dependency on payment-gateway.

Timeline (UTC)

14:20 — pay-gw-4.2.1 rollout begins, canary at 10%.

14:28 — Rollout reaches 100% of payment-gateway pods.

14:32 — First 503 responses observed on /v2/authorise.

14:38 — Error-rate alert pages payments-core-oncall.

14:44 — checkout-api circuit breaker opens, shedding load.

14:51 — Pool exhaustion identified from pgbouncer metrics.

14:58 — Decision taken to roll back pay-gw-4.2.1.

15:10 — Rollback complete, error rate returns to baseline.

Action items

• Add a pre-deploy check rejecting pool sizes below the documented minimum.

• Add a pgbouncer pool-saturation alert at 80% utilisation.

• Require load testing for any change to connection pool configuration.